Communications

Exam code: 0625 & 0972

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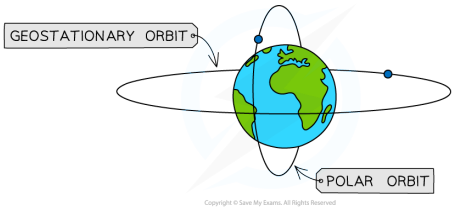
Communications with satellites

- Communications with **artificial satellites** use **microwaves**:
  - Some satellite phones use low-orbit artificial satellites
  - Some satellite phones and direct broadcast satellite televisions use geostationary satellites

Geostationary and polar-orbiting satellites



Geostationary and polar orbits around the Earth

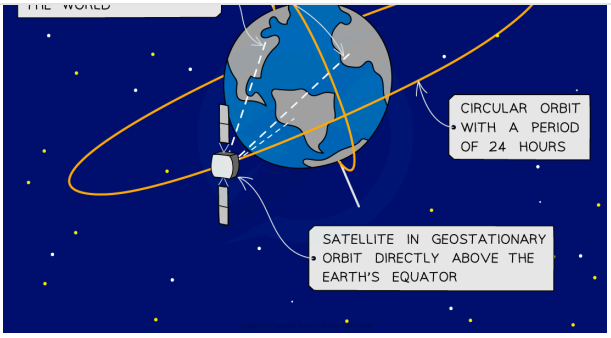
Low orbit satellites

- Low orbit satellites orbit much lower than geostationary satellites, at around 200 km above sea level
  - Used for monitoring the weather, military applications, and taking images of the Earth's surface
  - There is a much shorter time delay for signals compared to geostationary orbit signals
  - The signals and images are much clearer due to the lower orbit
  - However, there is limited use in any one orbit because more than one satellite is required for continuous operation
- Polar orbit satellites are a type of low Earth orbit satellite that orbit around the poles

Low orbit satellites

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

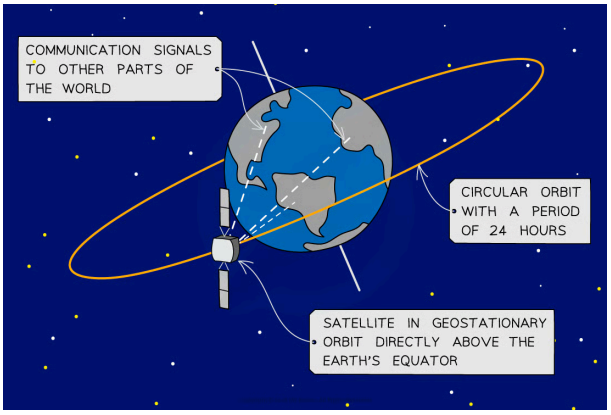


Some satellite phones use low-orbit artificial satellites if a more detailed signal is required

Geostationary satellites

- Geostationary satellites orbit **above the Earth's equator**
  - The orbit of the satellite is 24 hours
  - At a height of 36 000 km above the Earth's surface, much higher than polar satellites
  - Used for radio and telecommunication broadcasting around the world due to its high orbit

Orbit of geostationary satellite



Some satellite phones and direct broadcast satellite television use geostationary satellites

Systems of communications

Extended tier only

- Many important systems of communication rely on **long-wave** electromagnetic radiation, including:
  - Mobile phones, wireless internet & satellite television (using **microwaves**)
  - Bluetooth, terrestrial television signals & local radio stations (using **radio** waves)
  - Optical fibres (using **visible** or **infrared** waves)

distances
- As with radio waves, microwave signals will be clearer if there are no obstacles in the way which may cause **diffraction** of the beam
- On the ground, mobile phone signals use a network of **microwave transmitter masts** to relay the signals from the nearest mast to the receiving phone
    - They cannot be spaced so far apart that, for example, hills or the curvature of the Earth diffract the beam

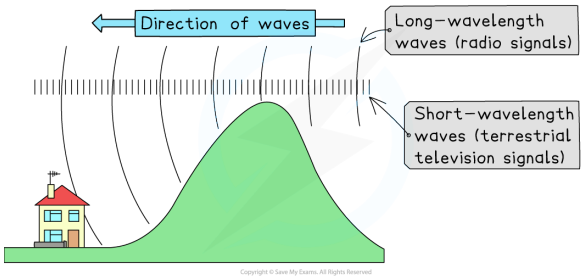
- Mobile phones and wireless internet use microwaves because microwaves are not refracted, reflected or absorbed by the **atmosphere** or **ionosphere**
    - This means satellites can relay signals around the Earth allowing communication 24 hours a day all around the world
    - Also, they can penetrate most walls and only require a short aerial for transmission and reception

Radio waves

- Radio waves can be used to transmit signals over **short** distances
- Radio waves have wavelengths of around a kilometre
- Radio waves can **diffract** around the hills
    - This is because radio waves have wavelengths similar to the size of the hill

- Bluetooth uses radio waves instead of wires or cables to transmit information between electronic devices, over short distances, such as phones and speakers
    - Bluetooth signals tend to have **shorter** wavelengths than radio or television signals
    - This enables high rates of data transmission that can only be used over a short distance (for example, within a household)
    - This means they can pass through walls but the signal is significantly **weakened** on doing so

Radio waves diffract around hills



Radio signals diffract around hills because they are a similar wavelength to the hill

Optical fibres

- Optical fibres use visible light and short-wavelength infrared radiation for cable television and high-speed

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